Metal and Acid Reactions

Aim: To observe the different reactivities of different metals and acids.

Equipment:

- 8 test tubes
- Test tube rack
- 2 corks or bungs to fit the test tubes loosely
- Bunsen burner
- Wood splint (1 per group)
- 25cm^3 dilute HCl (1M)
- 25cm^3 dilute sulphuric acid (0.5M)
- Granules, filiings of foil pieces of copper, iron, magnesium and zinc

Safety Information



Hydrochloric acid [1M] – possible bacterial presence

Wear safety goggles. Rinse skin and inform teacher immediately.

Sulphuric acid [0.5M] – irritant to skin and eye

Wear safety goggles. Rinse skin and inform teacher immediately.

Glassware – risk of cuts if broken.

Apply pressure to wounds and inform teacher immediately.

Method:

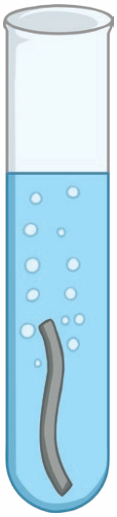
It might be useful to stand your test tube rack over plain white paper or a mini whiteboard to help you with labelling and identifying the contents.

Step 1: Place six test tubes in the test-tube rack.

Step 2: Add a 2–3cm depth of dilute hydrochloric acid to the first three tubes.

Step 3: Add a 2–3cm depth of dilute sulphuric acid to the remaining three tubes.

Step 4: Add a small piece of a different metal to each of the tubes with hydrochloric acid in them. Record which metal you add to each tube.



Step 5: Add a small piece of the same metals to each of the tubes with sulphuric acid in them. Record which metal you add to each tube.

Step 6: Observe any reactions and make a note of your observations in your table of results.

Step 7: From your observations, choose one of the metals that reacts rapidly with the acids. Collect and test the gas produced. Your teacher will show you how to test the gas being produced in these reactions.

Choose another of the metals that reacts rapidly with the acids. In a clean test tube, add a piece of this metal to a 2–3cm depth of one of the acids. This time, however, place a cork loosely in the top of the test tube so that any gas produced escapes slowly. Light a wood splint, remove the cork and immediately hold the flame to the mouth of the tube. If nothing happens, you may need to try again.

Results

Acid	Hydrochloric acid	Hydrochloric acid	Hydrochloric acid
Metal			
Observations			
Diagram			

Acid	Sulphuric acid	Sulphuric acid	Sulphuric acid
Metal			
Observations			
Diagram			

The gas indentified was: _____

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Questions

1. Write down the metals you tested in order of the most reactive to the least reactive.
2. What gas was produced by the reaction?
3. When a metal and an acid react, they always produce the gas you found and a salt. Show this as a word equation.
4. Describe how you tested for the gas, and the observations you made if the gas was present.
5. Complete the following word equations:
    a. hydrochloric acid + magnesium →
    b. hydrochloric acid + copper →
    c. sulphuric acid + iron →
    d. sulphuric acid + zinc →
6. Complete the symbol equation for the reaction between sulphuric acid and magnesium.
    a. H_2SO_4 + Mg →

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Answers

1. Write down the metals you tested in order of the most reactive to the least reactive.
   magnesium – zinc – iron - copper

2. What gas was produced by the reaction?
   hydrogen

3. When a metal and an acid react, they always produce the gas you found and a salt. Show this as a word equation.
   metal + acid → salt + hydrogen

4. Describe how you tested for the gas, and the observations you made if the gas was present.
   Capture the gas produced with a loosely fitting bung placed in the neck of the test tube, and test the gas using a lit splint held into it. If a squeaky pop is heard then hydrogen gas is present.

5. Complete the following word equations:

   a. hydrochloric acid + magnesium → magnesium chloride + hydrogen

   b. hydrochloric acid + copper → copper chloride + hydrogen

   c. sulphuric acid + iron → iron sulphate + hydrogen

   d. sulphuric acid + zinc → zinc sulphate + hydrogen

6. Complete the symbol equation for the reaction between sulphuric acid and magnesium.

   a. $H_2SO_4 + Mg \rightarrow MgSO_4 + H_2$